Instead, it spread. By the 1970s it was clear that so-called dark matter permeates most all clusters of galaxies and even individual galaxies. By the 2000s, it was clear that the dark matter gravitationally lenses light from more distant galaxies (Figure 24.3), just as Gargantua gravitationally lenses light from stars (Chapter 8). Today that lensing is being used to map the dark matter in our universe.



Fig. 24.3. Dark matter in the galaxy cluster Abell 2218 gravitationally lenses more distant galaxies. The images of the lensed galaxies are arc-shaped (e.g., those I circled in purple), analogous to arc structures seen in Gargantua's gravitational lensing, Chapter 8.

And today physicists are fairly sure that the dark matter is truly revolutionary, that it consists of fundamental particles of a type never before seen, but a type predicted by our best current understanding of the quantum laws of physics. Physicists have embarked on a holy-grail mission: a quest to detect these particles of dark matter, shooting through the Earth with near impunity, and measure their properties.